

SQL Queries Used in Car Rental System Project

1. List all available cars

```
SELECT * FROM cars WHERE Available > 0;
```

2. List all client users (non-admins)

```
SELECT * FROM users WHERE Type = 0;
```

3. Count the total number of cars

```
SELECT COUNT(*) AS TotalCars FROM cars;
```

4. Total rentals done by each user

```
SELECT u.FirstName, u.LastName, COUNT(r.ID) AS Rentals  
FROM rents r  
JOIN users u ON r.User = u.ID  
GROUP BY r.User;
```

5. Car models rented more than once

```
SELECT c.Model, COUNT(r.ID) AS TimesRented  
FROM rents r  
JOIN cars c ON r.Car = c.ID  
GROUP BY r.Car  
HAVING COUNT(r.ID) > 1;
```

6. Update availability for a specific car

```
UPDATE cars SET Available = Available - 1 WHERE ID = 8;
```

7. Delete a rent entry

```
DELETE FROM rents WHERE ID = 9;
```

8. Average rental duration

```
SELECT AVG(Hours) AS AvgHours FROM rents;
```

9. Total revenue per car

```
SELECT c.Model, SUM(r.Total) AS Revenue  
FROM rents r  
JOIN cars c ON r.Car = c.ID  
GROUP BY r.Car;
```

10. Latest 5 rental entries

```
SELECT * FROM rents ORDER BY DateTime DESC LIMIT 5;
```

11. Users who rented cars in 2023

```
SELECT DISTINCT u.FirstName, u.Email
FROM users u
JOIN rents r ON u.ID = r.User
WHERE r.DateTime LIKE '2023%';
```

12. Cars manufactured after 2025

```
SELECT * FROM cars WHERE Year > 2025;
```

13. Rentals longer than 5 hours

```
SELECT * FROM rents WHERE Hours > 5;
```

14. Users who spent more than 5000

```
SELECT ID, FirstName
FROM users
WHERE ID IN (
    SELECT User FROM rents GROUP BY User HAVING SUM(Total) > 5000
);
```

15. Increase price of cars after 2025 by 10%

```
UPDATE cars SET Price = Price * 1.10 WHERE Year > 2025;
```

16. Rental details with car and user info

```
SELECT u.FirstName, u.LastName, c.Model, c.Brand, r.DateTime, r.Total
FROM rents r
JOIN users u ON r.User = u.ID
JOIN cars c ON r.Car = c.ID;
```

17. Unique users per car rented

```
SELECT c.Model, COUNT(DISTINCT r.User) AS UniqueRenters
FROM rents r
JOIN cars c ON r.Car = c.ID
GROUP BY r.Car;
```

18. Users who never rented a car

```
SELECT u.ID, u.FirstName, u.Email
FROM users u
LEFT JOIN rents r ON u.ID = r.User
WHERE r.ID IS NULL;
```

19. Car availability status display

```
SELECT Brand,
    CASE WHEN Available > 0 THEN 'Yes' ELSE 'No' END AS IsAvailable
FROM cars;
```

20. List latest rentals with car name

```
SELECT r.ID, u.FirstName, c.Model, r.DateTime  
FROM rents r  
JOIN users u ON r.User = u.ID  
JOIN cars c ON r.Car = c.ID  
ORDER BY r.DateTime DESC;
```